Ahmedabad University

GROUP 2: CLIMATE

Probabilistic Forecasting of Extreme Weather Events

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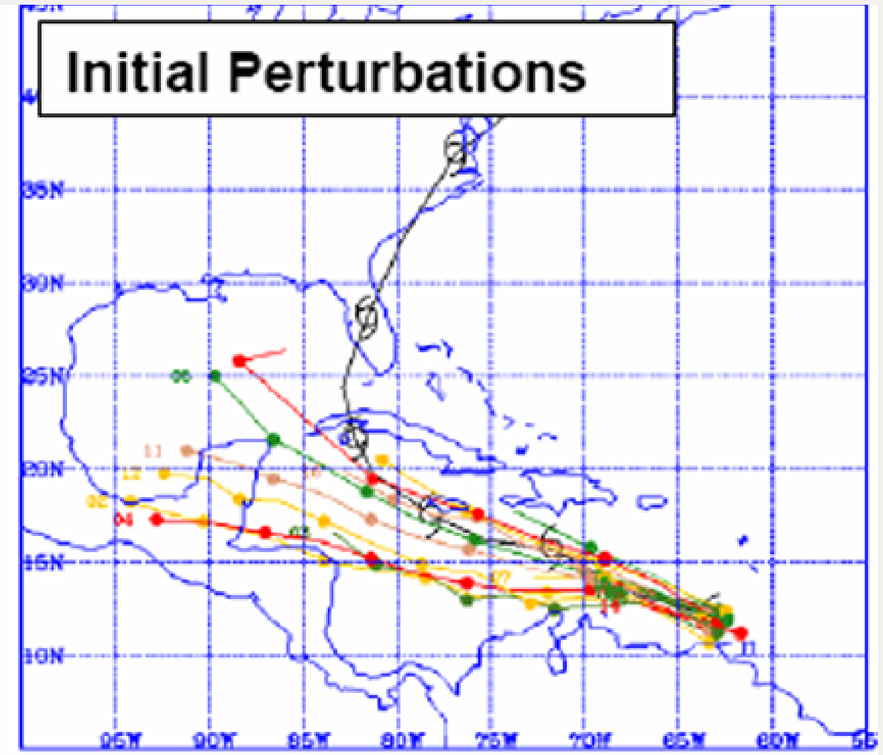
Project Context and Motivation

Title: Probabilistic Modeling of Extreme Weather Events for Risk-Informed Decision Support

Application Domain: Defense operations, infrastructure resilience, and weather risk assessment

Context: Weather forecasts are critical for operational planning and infrastructure protection.

Atmospheric systems are chaotic and highly sensitive to initial conditions and model assumptions, making uncertainty unavoidable—especially for rare and high-impact events.



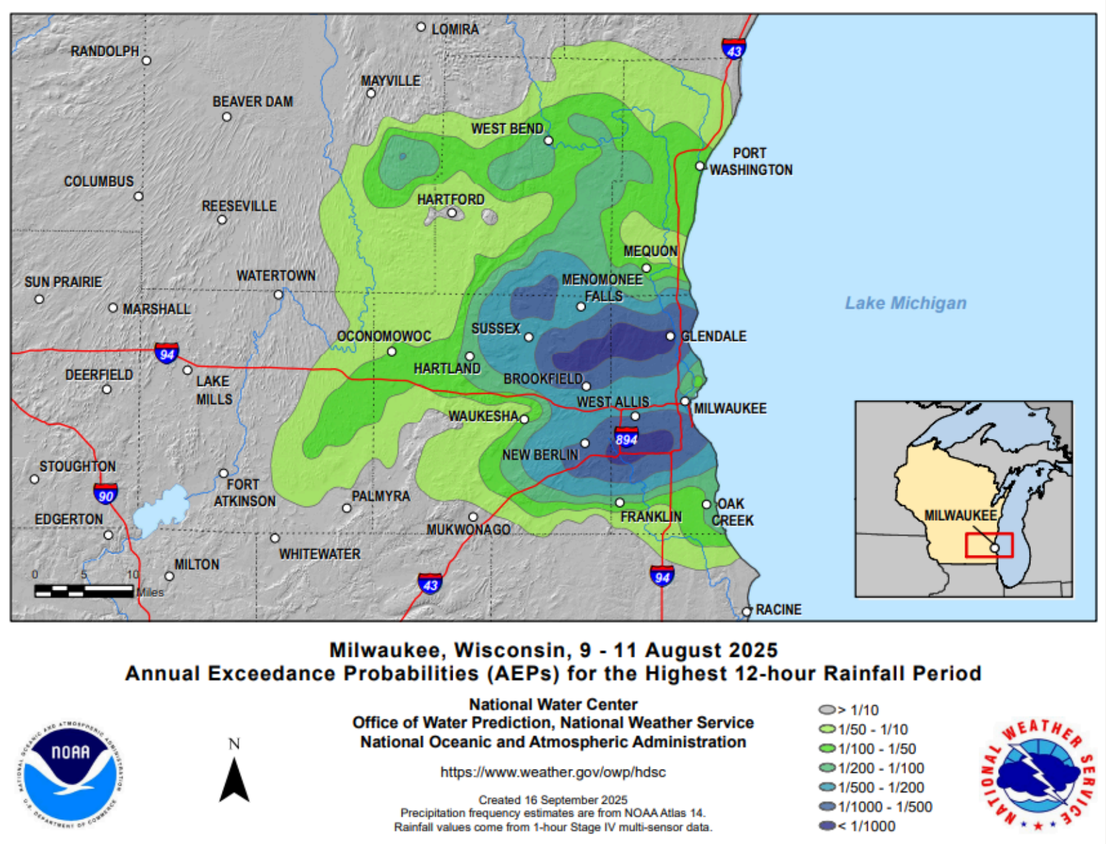
https://www.weather.gov/owp/hdsc_aep

Project Objective and Scope

Objective: To quantify uncertainty in extreme weather events using probabilistic methods.

Scope:

- Use ensemble forecasting to generate multiple plausible weather outcomes
- Apply Bayesian inference to estimate event likelihoods
- Provide probabilistic risk estimates rather than deterministic predictions



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Project System Overview

End to End Pipeline:

- Generate ensemble weather forecasts
- Extract relevant extreme-event indicators
- Model uncertainty probabilistically
- Apply Bayesian inference for likelihood estimation

Major Components:

- Ensemble forecast data
- Probabilistic model
- Bayesian inference module

Sources of Uncertainty

- **Initial Condition Uncertainty**: Imperfect observations of the atmosphere
- **Model Uncertainty:** Approximations in physical parameterizations
- **Environmental Variability:** Natural randomness in weather evolution
- **Data Uncertainty:** Limited or noisy historical observations of extreme events

Key Random Variables and Their Roles

- **Atmospheric State Variables (e.g., rainfall, wind speed)**:
- **Ensemble Forecast Outputs**
- **Extreme Event Indicator**
- **Model Uncertainty Term**
- **Prior Event Probability**

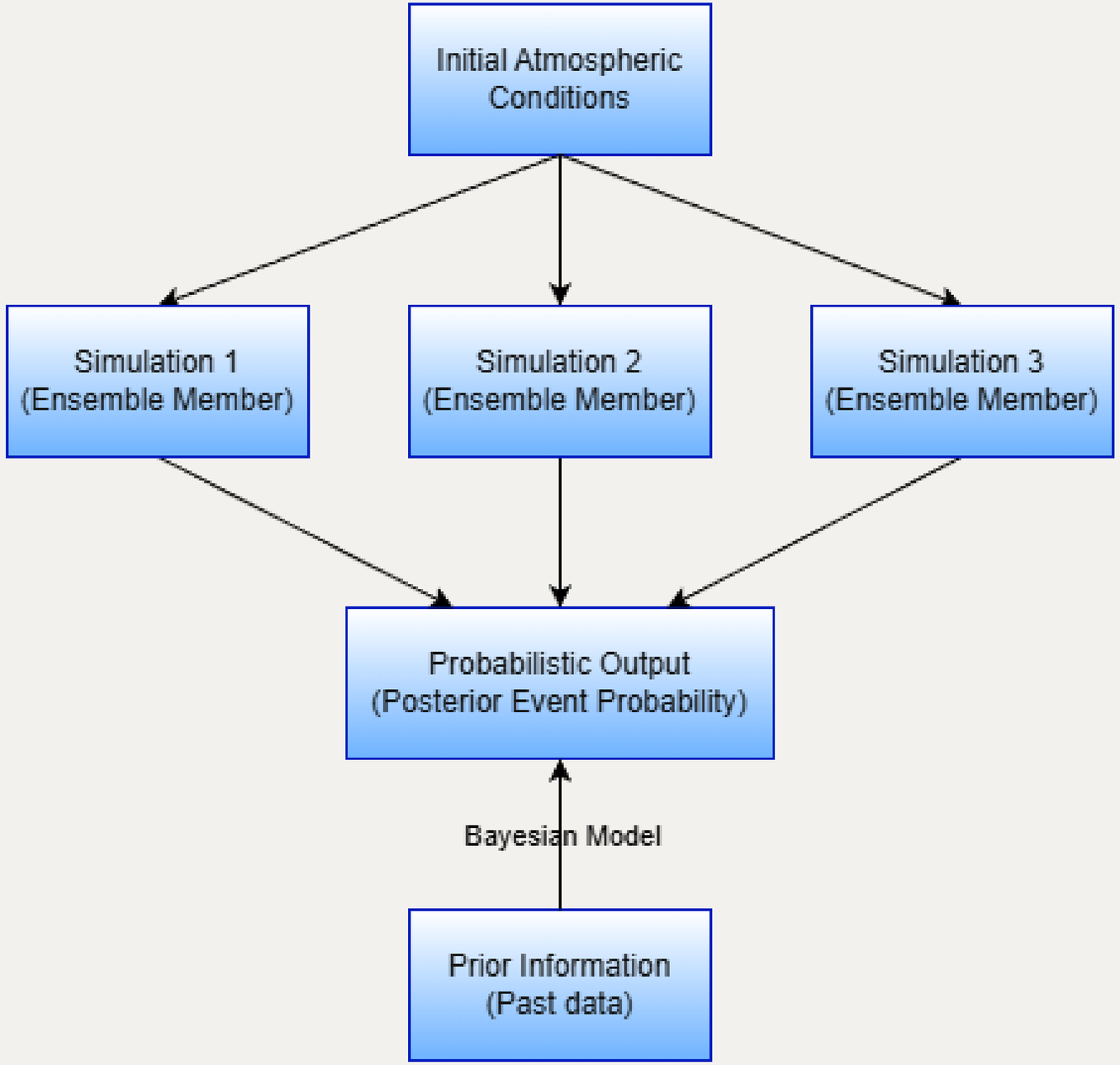
Probabilistic Models and Assumptions

Ensemble Weather Forecasting

- Approximate the uncertainty via multiple ensemble members

Assumptions

- Conditionally independent members
- Bayesian updating using past data

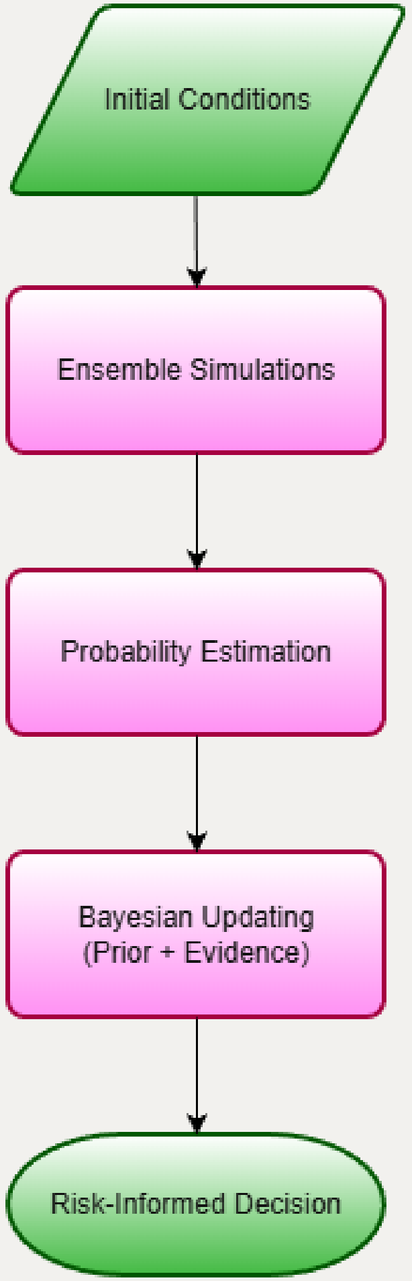


Probabilistic Reasoning and Inference Logic

- Uncertainty Propagation
- Ensemble Variability
- Conditional Inference
- Probability Aggregation
- Risk-Based Decision Support

Operationalizing Probabilistic Reasoning in the Project

- Available ensemble forecast data are considered probabilistic samples.
- Ensemble outputs are interpreted with Bayesian reasoning.
- Performance is evaluated using probability based metrics.
- Ensemble models represent uncertainty, rather than giving a deterministic value



Current Limitations and Conceptual Gaps

Component	Current Limitation/Gap
Initial condition perturbations	Need Refinement
Model Uncertainty	Not fully captured
Prior distributions	Data-limited
Independence assumption	Correlation possible
Ensemble size	Maybe Insufficient

Planned Refinements and Role Coordination

Next Milestone Refinements:

- Formalize random variables and distributions
- Introduce Bayesian updating for probability estimates

Conceptual and Modelling Tasks:

- Define probabilistic thresholds for extreme events
- Choose suitable measures to evaluate probability accuracy.

Role Coordination:

- **Github Manager:** Keep model organized and consistent.
- **Scribes:** Note changes and update assumptions
- **Video Team:** Explain how system works and how uncertainty is handled

Summary of Group-Level Understanding

Probabilistic Model

- Ensemble simulations represent uncertainty
- Extreme events are estimated using probability

Inference Logic

- Bayesian updating combines prior + evidence
- Conditional independence assumed for members

Project Direction

- Integrate model uncertainty in next milestone
- Evaluate using probability based metrics

THANK YOU!